An IMDb Dashboard

Lecture 2: Streamlit and SQL aggregates

Gregory S. DeLozier, PhD

Today

Layer	Responsibility
PostgreSQL	Store rows and execute SQL
database.py	Own queries and connections
app_basics.py	Controls, tables, charts

Exercise: Resume and Verify

```
bash start-postgres.sh
source .venv/bin/activate
export DATABASE_URL=postgresql://student@127.0.0.1:5433/imdb
python check_setup.py
```

Expected: imported tables and indexes are still present.

Exercise: Open the Starter

```
streamlit run app_basics.py --server.address 0.0.0.0
```

Open web port 8501. Leave database port 5433 local.

Starter: title search and yearly summaries.

Full app: Search, Notes, and graphical Reports.

One Parameterized Query

```
connection.execute(
    "SELECT primary_title FROM imdb_course.titles "
    "WHERE start_year = %s",
    (1994,),
)
```

SQL text and parameter values travel separately.

Connection and Display Lifecycle

Event	What happens
Query call	Open a connection; execute; close
Interface rerun	Reuse cached results when available
Refresh button	Clear the result cache

Cache rows; keep connection ownership simple.

A Join Produces Movie Rows

```
SELECT t.primary_title, t.start_year,
       r.average_rating, r.num_votes
FROM imdb_course.titles t
JOIN imdb_course.ratings r USING (title_id)
WHERE r.num_votes >= 1000;
```

The inner join leaves out unrated titles.

Grouping Produces Yearly Rows

```
SELECT t.start_year, count(*) AS movies,
       round(avg(r.average_rating), 2) AS mean_rating
FROM imdb_course.titles t
JOIN imdb_course.ratings r USING (title_id)
WHERE t.start_year IS NOT NULL AND r.num_votes >= 1000
GROUP BY t.start_year
ORDER BY t.start_year;
```

What the Dashboard Measures

Output	Meaning
movies	Qualifying titles in this sample
mean_rating	Equal-weight mean of title ratings
start_year	Reported release year

Source order and filtering affect the chart.

WHERE and HAVING

```
-- Keep rows before grouping.
WHERE r.num_votes >= 1000

-- Keep groups after aggregation.
HAVING count(*) >= 5
```

A group threshold belongs after GROUP BY.

Exercise: A Runtime Control

Files: `database.py`, `app_basics.py`

Add a minimum runtime to `yearly_summary`.

Pass it from a Streamlit number input and include it in the cache key.

Expected: short movies leave the yearly totals.

The Working Boundary

SQL determines which rows and summaries mean something.

The data layer returns them.

Streamlit makes the result inspectable.

Next: rank and compare rows without collapsing them.